

name, might be Egyptian and might not actually be associated with peasants. We continue by covering Euclid’s algorithm, an important “classic” algorithm for finding greatest common divisors. Finally, we cover an algorithm from Japan that generates magic squares.

Russian Peasant Multiplication

Many people remember learning the multiplication table as a particularly painful part of their education. Young children ask their parents why learning the multiplication table is necessary, and parents usually respond that they can’t multiply without knowing it. How wrong they are. *Russian peasant multiplication(RPM)* is a method that enables people to multiply large numbers without knowing most of the multiplication table.

RPM’s origins are unclear. An ancient Egyptian scroll called the Rhind papyrus contains a version of this algorithm, and some historians have proposed (mostly unconvincing) conjectures about how the method could have spread from ancient Egyptian scholars to the peasants of the vast Russian hinterlands. Regardless of the details of its history, RPM is an interesting algorithm.

Doing RPM by Hand

Consider the task of multiplying 89 by 18. Russian peasant multiplication proceeds as follows. First, create two columns next to each other. The first column is called the *halving* column and starts with 89. The second column is the *doubling* column and starts with 18 ([Table 2-1](#)).

[Table 2-1:](#) *Halving/Doubling Table, Part 1*

Halving	Doubling
89	18

We’ll fill out the halving column first. Each row of the halving column takes the previous entry and divides it by 2, ignoring the remainder. For example, 89 divided by 2 is 44 remainder 1, so we write 44 in the second row of the halving column ([Table 2-2](#)).